

Import Libraries :

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Load Dataset :

```
df = pd.read_csv("/content/sample_data/ecommerce_customer_behavior_dataset.csv")
```

Data Analysis :

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17049 entries, 0 to 17048
Data columns (total 17 columns):
 #   Column                                Non-Null Count  Dtype  
---  -
 0   Customer_ID                          17049 non-null  object 
 1   Date                                 17049 non-null  object 
 2   Month                               17049 non-null  object 
 3   Age                                  17049 non-null  int64  
 4   Gender                              17049 non-null  object 
 5   City                                 17049 non-null  object 
 6   Product_Category                    17049 non-null  object 
 7   Unit_Price                          17049 non-null  float64 
 8   Quantity                            17049 non-null  int64  
 9   Discount_Amount                     17049 non-null  float64 
10   Total_Amount                        17049 non-null  float64 
11   Payment_Method                      17049 non-null  object 
12   Device_Type                         17049 non-null  object 
13   Session_Duration_Minutes            17049 non-null  int64  
14   Is_Returning_Customer               17049 non-null  bool    
15   Delivery_Time_Days                  17049 non-null  int64  
16   Customer_Rating                     17049 non-null  int64  
dtypes: bool(1), float64(3), int64(5), object(8)
memory usage: 2.1+ MB
```

```
rs.get_loc("Date") + 1, "Month", pd.to_datetime(df["Date"]).dt.month_name())
```

```
df.describe()
```


	Order_ID	Customer_ID	Date	Age	Gender	City	Product_Category
0	ORD_000001-1	CUST_00001	2023-05-29	40	Male	Ankara	Books
1	ORD_000001-2	CUST_00001	2023-10-12	40	Male	Ankara	Home & Garden
2	ORD_000001-3	CUST_00001	2023-12-05	40	Male	Ankara	Sports
3	ORD_000002-1	CUST_00002	2023-05-11	33	Male	Istanbul	Food
4	ORD_000002-2	CUST_00002	2023-06-16	33	Male	Istanbul	Beauty
5	ORD_000003-1	CUST_00003	2023-02-27	42	Male	Konya	Toys
6	ORD_000003-2	CUST_00003	2024-01-03	42	Male	Konya	Home & Garden
7	ORD_000004-1	CUST_00004	2024-02-13	53	Male	Izmir	Fashion
8	ORD_000005-1	CUST_00005	2023-03-16	32	Male	Ankara	Home & Garden
9	ORD_000005-2	CUST_00005	2023-06-12	32	Male	Ankara	Toys

```
df.drop(columns={"Pages_Viewed"}, axis = 1, inplace= True)
```

```
df.columns
```

```
Index(['Order_ID', 'Customer_ID', 'Date', 'Month', 'Age', 'Gender',
      'City',
      'Product_Category', 'Unit_Price', 'Quantity',
      'Discount_Amount',
      'Total_Amount', 'Payment_Method', 'Device_Type',
      'Session_Duration_Minutes', 'Pages_Viewed',
      'Is_Returning_Customer',
      'Delivery_Time_Days', 'Customer_Rating'],
      dtype='object')
```

```
pd.isnull(df).sum()
```

	0
Customer_ID	0
Date	0
Month	0
Age	0
Gender	0
City	0
Product_Category	0
Unit_Price	0
Quantity	0
Discount_Amount	0
Total_Amount	0
Payment_Method	0
Device_Type	0
Session_Duration_Minutes	0
Is_Returning_Customer	0
Delivery_Time_Days	0
Customer_Rating	0

dtype: int64

`df.columns`

```
Index(['Customer_ID', 'Date', 'Month', 'Age', 'Gender', 'City',
      'Product_Category', 'Unit_Price', 'Quantity',
      'Discount_Amount',
      'Total_Amount', 'Payment_Method', 'Device_Type',
      'Session_Duration_Minutes', 'Is_Returning_Customer',
      'Delivery_Time_Days', 'Customer_Rating'],
      dtype='object')
```

```
df.drop(columns= ["Order_ID"], axis = 1,inplace = True)
```

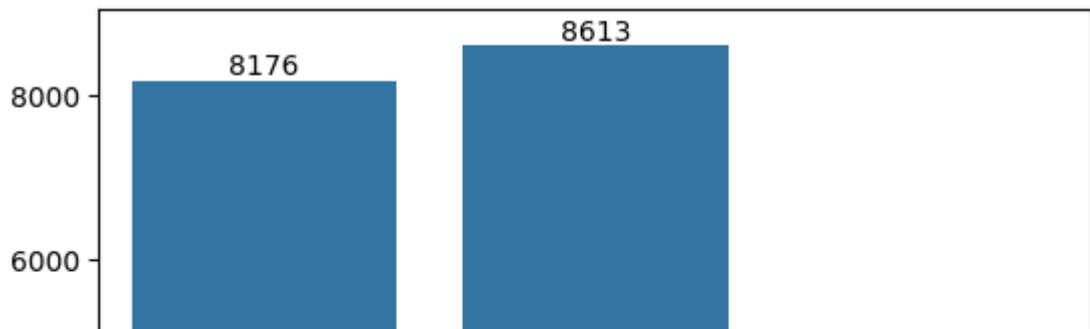
`df.columns`

```
Index(['Customer_ID', 'Date', 'Month', 'Age', 'Gender', 'City',
      'Product_Category', 'Unit_Price', 'Quantity',
      'Discount_Amount',
      'Total_Amount', 'Payment_Method', 'Device_Type',
      'Session_Duration_Minutes', 'Is_Returning_Customer',
      'Delivery_Time_Days', 'Customer_Rating'],
      dtype='object')
```

Exploratory Data Analysis:

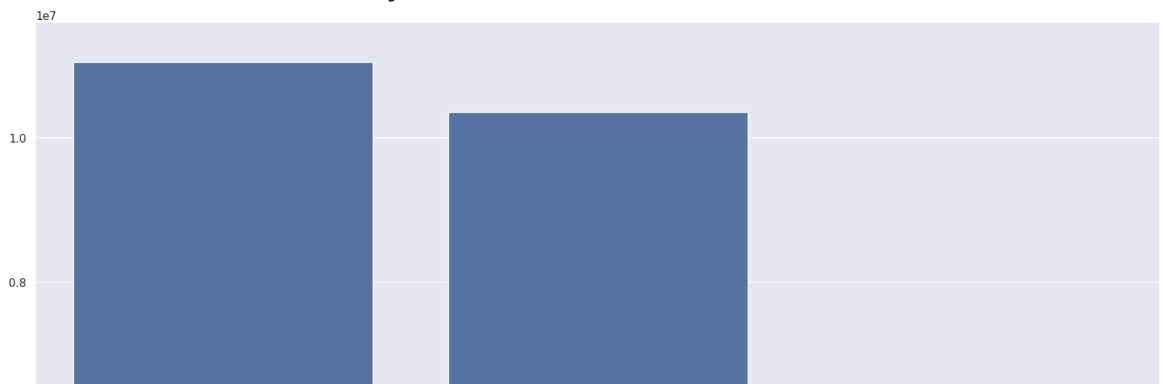
Gender:

```
ax = sns.countplot(data=df,x="Gender")
for bars in ax.containers:
    ax.bar_label(bars)
```



```
sns.set(rc={'figure.figsize' : (20,15)})
sns.rc=setsales_gen = df.groupby(['Gender'],as_index=False)['Total_Amount'].
sns.barplot(data=sales_gen,x ='Gender',y= 'Total_Amount')
```

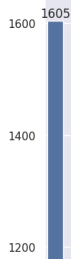
<Axes: xlabel='Gender', ylabel='Total_Amount'>



So,Females are highest purchasing customers

Age:

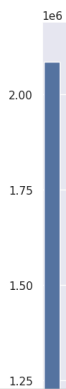
```
sns.set(rc={'figure.figsize' : (20,17)})
ax = sns.countplot(data=df,x="Age")
for bars in ax.containers:
    ax.bar_label(bars)
```



1605
1600
1400
1200

```
sns.set(rc={'figure.figsize' : (20,15)})  
sales_age = df.groupby(['Age'],as_index=False)['Total_Amount'].sum().sort_va  
sns.barplot(data=sales_age,x = 'Age',y= 'Total_Amount')
```

<Axes: xlabel='Age', ylabel='Total_Amount'>

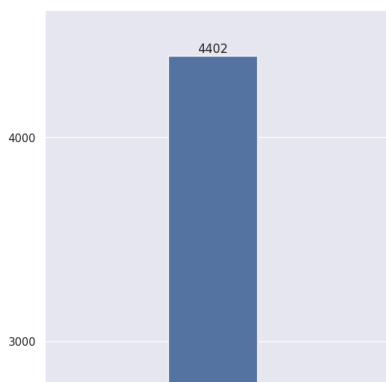


1e6
2.00
1.75
1.50
1.25

Highest purchasing age range is 25-45

City:

```
sns.set(rc={'figure.figsize' : (20,17)})  
ax = sns.countplot(data=df,x="City")  
for bars in ax.containers:  
    ax.bar_label(bars)
```



4402
4000
3000

```
sns.set(rc={'figure.figsize' : (20,15)})  
sales_city = df.groupby(['City'],as_index=False)['Total_Amount'].sum().sort_  
sns.barplot(data=sales_city,x = 'City',y= 'Total_Amount')
```

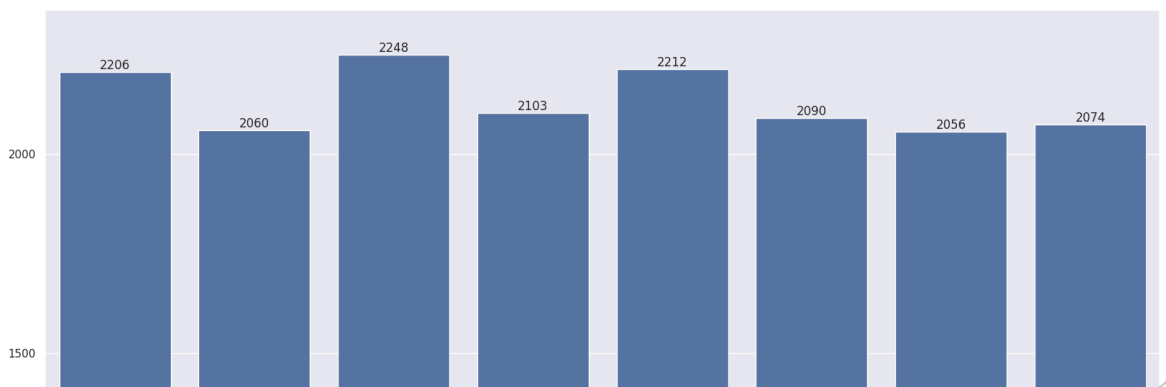
<Axes: xlabel='City', ylabel='Total_Amount'>



Istanbul, Ankara and Izmir are the highest purchasing cities

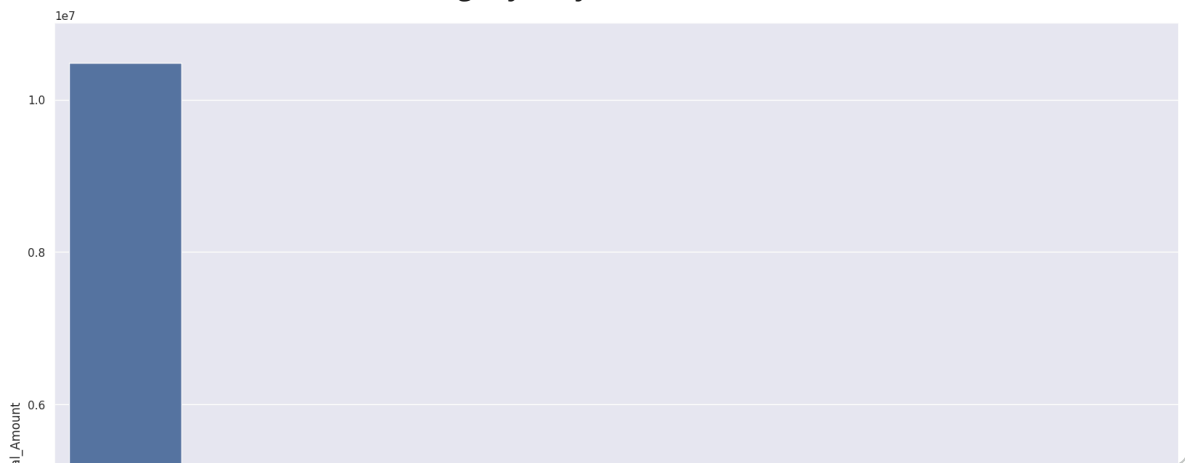
Product Category:

```
sns.set(rc={'figure.figsize' : (20,17)})  
ax = sns.countplot(data=df,x="Product_Category")  
for bars in ax.containers:  
    ax.bar_label(bars)
```



```
sns.set(rc={'figure.figsize' : (20,15)})  
sales_product = df.groupby(['Product_Category'],as_index=False)['Total_Amount']  
sns.barplot(data=sales_product,x = 'Product_Category',y= 'Total_Amount')
```

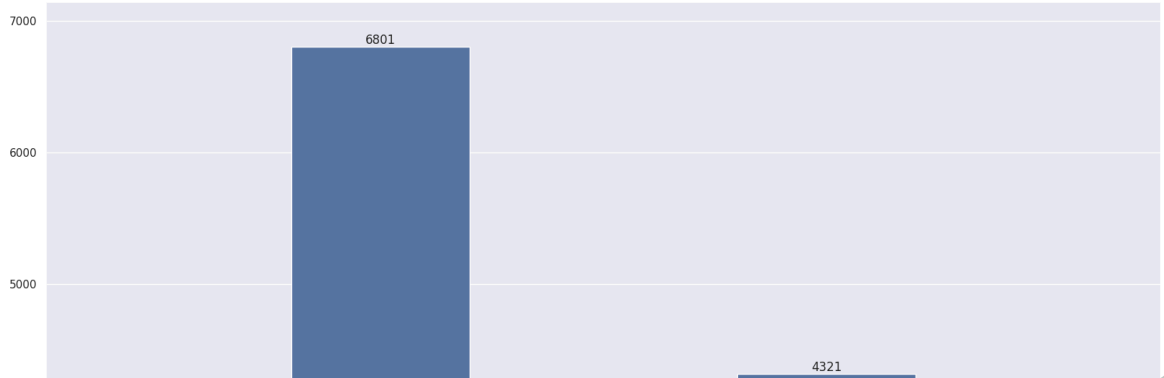
<Axes: xlabel='Product_Category', ylabel='Total_Amount'>



Sports, Beauty, and Books lead in purchase volume, while Electronics and Home & Garden generate the highest sales revenue.

Payment_Method:

```
sns.set(rc={'figure.figsize' : (20,17)})
ax = sns.countplot(data=df,x="Payment_Method")
for bars in ax.containers:
    ax.bar_label(bars)
```



```
sns.set(rc={'figure.figsize' : (20,15)})
sales_pay = df.groupby(['Payment_Method'],as_index=False)['Total_Amount'].sum()
sns.barplot(data=sales_pay,x = 'Payment_Method',y= 'Total_Amount')
```

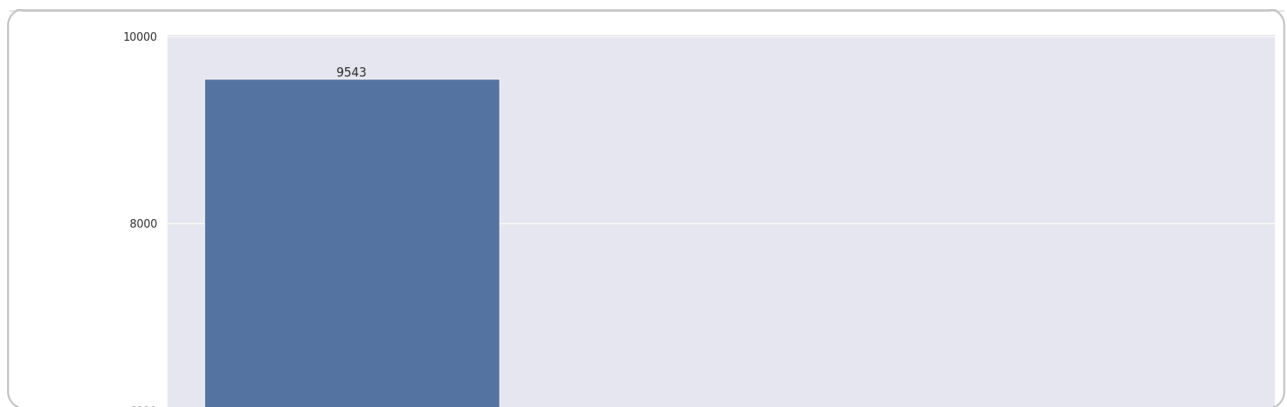
<Axes: xlabel='Payment_Method', ylabel='Total_Amount'>



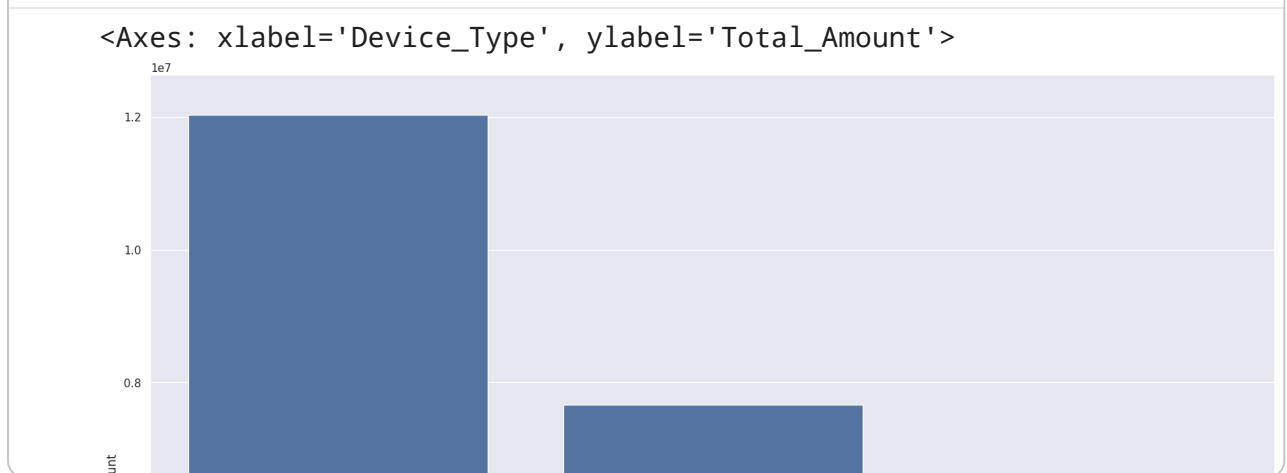
Credit Card and Debit Card are the leading payment methods in terms of both transaction volume and total sales value.

Device Type:

```
sns.set(rc={'figure.figsize' : (20,17)})
ax = sns.countplot(data=df,x="Device_Type")
for bars in ax.containers:
    ax.bar_label(bars)
```

```
sns.set(rc={'figure.figsize' : (20,15)})
sales_device = df.groupby(['Device_Type'],as_index=False)['Total_Amount'].sum()
sns.barplot(data=sales_device,x = 'Device_Type',y= 'Total_Amount')
```



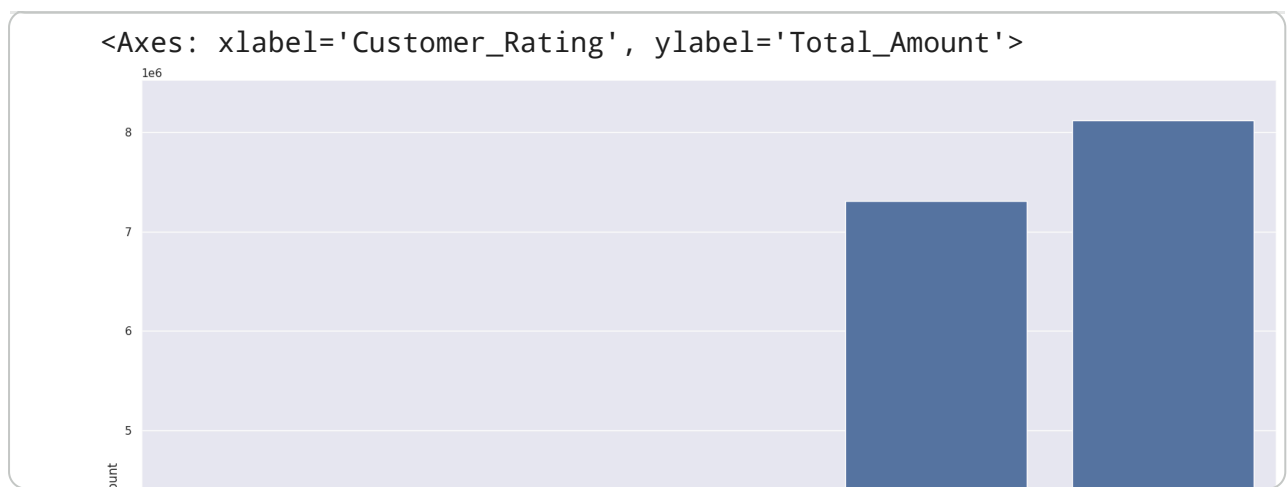
Mobile is used to purchase

Customer Rating:

```
sns.set(rc={'figure.figsize' : (20,17)})
ax = sns.countplot(data=df,x="Customer_Rating")
for bars in ax.containers:
    ax.bar_label(bars)
```

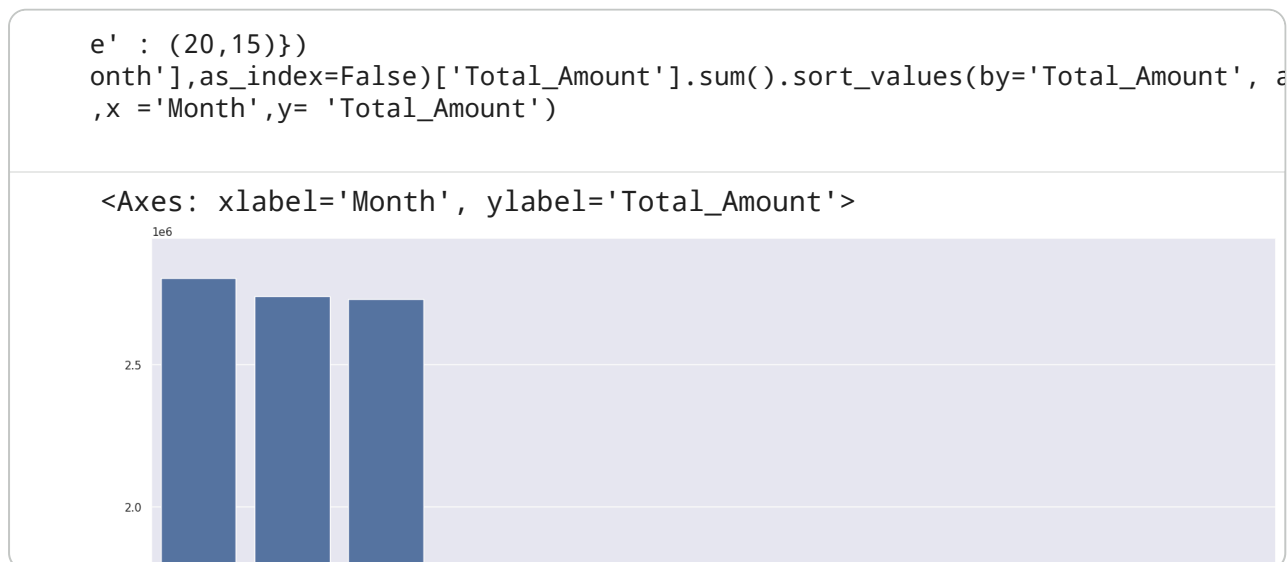
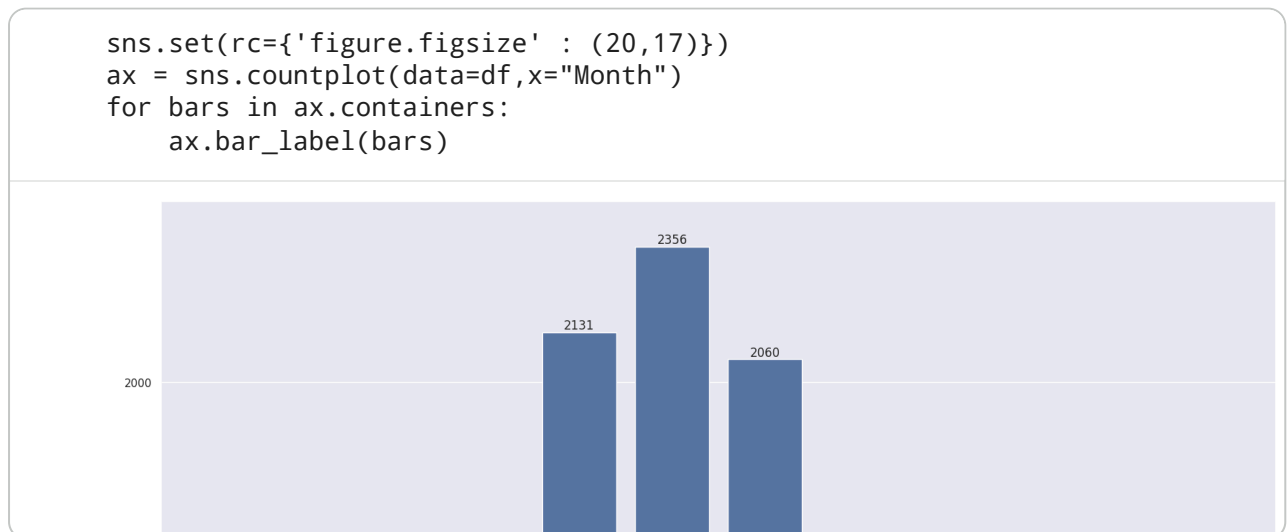


```
sns.set(rc={'figure.figsize' : (20,15)})
sales_custrat = df.groupby(['Customer_Rating'],as_index=False)['Total_Amount'].sum()
sns.barplot(data=sales_custrat,x = 'Customer_Rating',y= 'Total_Amount')
```



Most customers gave a 5-star rating, indicating a high level of customer satisfaction.

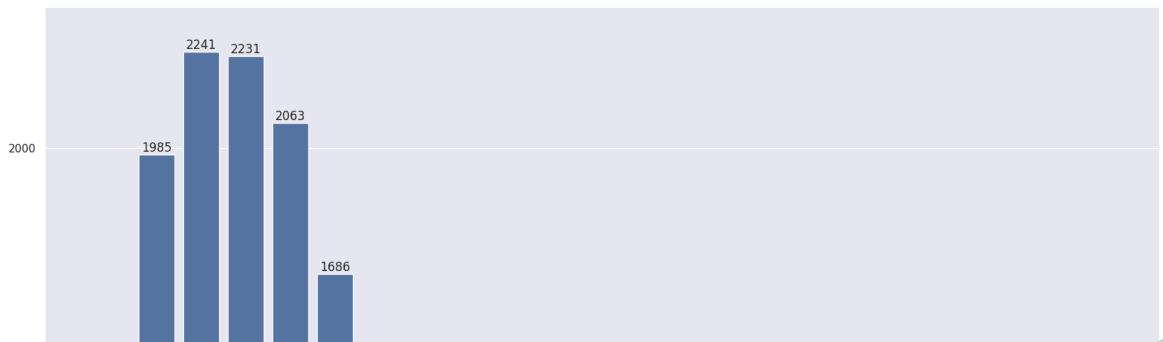
Month:



January, February, and March recorded the highest product purchases, both in transaction count and total amount.

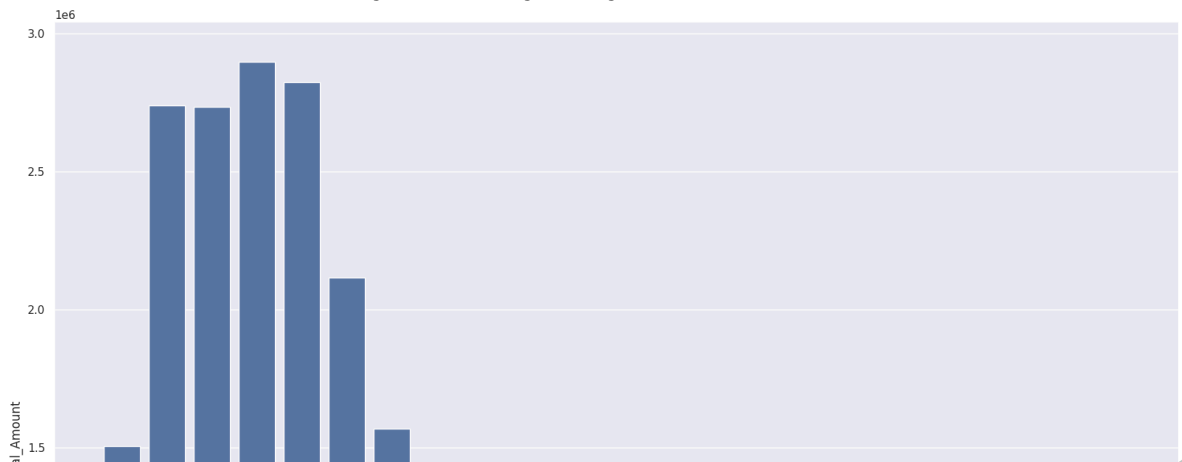
Delivery Time Days:

```
sns.set(rc={'figure.figsize' : (20,17)})
ax = sns.countplot(data=df,x="Delivery_Time_Days")
for bars in ax.containers:
    ax.bar_label(bars)
```



```
False)['Total_Amount'].sum().sort_values(by='Total_Amount', ascending=False)
['Total_Amount']
```

<Axes: xlabel='Delivery_Time_Days', ylabel='Total_Amount'>



Most orders are delivered within 4–5 days.

Conclusion:

The analysis of 17,049 e-commerce transactions shows that the business is performing well, with high customer satisfaction and strong sales across multiple product categories. Female customers and the 25–45 age group contribute the most to sales. Istanbul, Ankara, and Izmir are the top-performing cities, while Electronics and Home & Garden generate the highest revenue despite Sports, Beauty, and Books having the highest purchase volume. January, February, and March recorded the highest sales performance, leading in both transaction volume and total revenue. Most customers prefer Credit Card and Debit Card payments, mobile devices are the primary shopping platform, customers largely give 5-star ratings, and most orders are delivered within 4–5 days. These findings indicate a healthy customer experience and a strong market position.

Advice :

1. Increase inventory and marketing for Electronics and Home & Garden to